The EDIBLES survey

IX: Simulations of the λ6614 DIB profile variations: a surprising connection with CH^+

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Abstract

The profiles of several diffuse interstellar bands (DIBs) show substructures that resemble unresolved rotational bands of electronic transitions of large molecules. Their profiles show clear variations along lines of sight probing different physical conditions. Analysis of variations in such profiles can constrain the sizes and geometries of the DIB carriers and the physical conditions of the interstellar environments in which they reside. We investigate the properties of rotational band contours for perpendicular transitions in planar, oblate symmetric top molecules and compare such contours to the observed profile structure of the λ6614 DIB. We investigate the shapes of the profiles as a function of the model parameters: the rotational constant B in the ground state, the relative change in the rotational constant of the excited state ΔB, the rotational excitation temperature $T_{\rm rot}$ and the Coriolis coupling constant ζ. We determine which parameters can reproduce the overall triple-peak profile of the λ6614 DIB as well as the variations across different lines of sight. We can reproduce the substructures in the λ6614 DIB with an oblate top with rotational constant $B = (2.2 \pm 1.8) \times 10^{-3}$ cm^{-1}, $\Delta B = (-7.2 \pm 0.4) \times 10^{-2}$%, and Coriolis coupling constant $\zeta = (2.9 \pm 0.1) \times 10^{-1}$ cm^{-1}. Thus, if the λ6614 DIB carrier conforms to an oblate symmetric top geometry, it is most likely to be a ~54 C atom molecule. The profile variations correspond to changes in the rotational temperature from 81 to 93 K. We furthermore find that the intrinsic line width is a key parameter for each sightline, and requires a range from 0.14 to 0.21cm^{-1} across our sample to reproduce the observations. The intrinsic line width of the λ6614 DIB correlates with the width of the CH^+ lines, suggesting an origin in the same environment. We conclude that the λ6614 DIB carrier resides in the same hot gas at low density that is probed by CH^+, and could chemically be affected by vibrationally excited H_2.

1 Introduction

The diffuse interstellar bands (DIBs) are a set of more than 500 absorption features from the near-UV to the near-IR range of the spectrum which are likely caused by molecules that reside in the Interstellar Medium (ISM; see e.g. Herbig [1], Sarre [2] for reviews; DIB [3] for a compilation of

relevant research and Hobbs et al.[4,5], Fan et al.[6] for DIB surveys). With just one exception, the precise molecular identity of the DIBs remains unknown. The DIBs were recorded for the first time by Heger[7] and their interstellar origin was established later on by Merrill[8]. DIBs have been found to be ubiquitous throughout the Universe; they have been observed within our Galaxy[9] and studied in extragalactic regions as well[10–14], which implies that DIB carriers must come from abundant species that are stable enough to survive the harsh environment (e.g UV radiation) of the ISM. Thus, the current consensus is that the DIB carriers are large carbonaceous molecules in the gas phase such as carbon chains, polycyclic aromatic hydrocarbons (PAHs), and fullerenes and their derivatives see e.g.[15–22]. So far, only C_{60}^{+} has been clearly associated with 5 DIBs in the infrared[23,24].

DIBs come in a wide variety of widths, strengths and band profile shapes. The DIB widths (measured as the full width at half maximum, FHWM) vary from 0.5 to 30 Å with a recent discovery of a new DIB at 7700 Åwith FWHM of 177 Å[25]. However, majority of DIBs have a width between 0.5 and 3 Å[5]. The strongest DIB λ4428 is symmetric in nature and has a broad Lorentzian profile, suggesting an origin in lifetime broadening due to a short-lived excited state[26]. On the other hand, some of the narrower DIBs such as $\lambda\lambda$5797, 6379 and 6614 show substructure and asymmetries at high spectral resolution. These observed double and triple peak structures are likely unresolved rotational contours of large molecules see e.g.[18,20,22,27,28]. As an alternative explanation Webster[29] proposed that the substructures are due to isotopic substitution in large carbonaceous molecules, where some of the ^{12}C atoms in the molecules are replaced by its isotope ^{13}C. In that case, each of the substructure peaks arises from the same transition from an isotopologue with a different number of ^{13}C atoms. However, isotopic substitution cannot explain the observed variations in the precise position of the substructure peaks seen across sightlines ([20,22]; see also Sect. 4.2).

If the observed substructures in the DIB line profiles are indeed due to the rotational contours of large molecules, we can gain more insight into the nature of the carrier molecules from detailed analyses of the line profiles, and from simulating the band contours that are expected for molecular

carriers of different sizes and geometries. Cossart-Magos and Leach[27] simulated vibronic transitions of selected PAH molecules (such as coronene) at temperatures between 3 and 100K and compared them to the observed profiles of four narrow DIBs. They argue that PAHs should indeed show clear structures in their band profiles, whereas such structures are not seen for most of the strong DIBs. Edwards and Leach[28] carried out similar calculations for fullerenes, adopting a rotational temperature of 100 K, producing similar sub structures in the simulated profiles. Schulz et al.[30] showed that linear cumulene carbene molecules (C_nH_2) can produce substructures that are comparable to some of the observed DIB profiles.

Of particular interest is the profile of the λ6614 DIB. This is a fairly strong DIB, and its profile shows a clear triple-peak substructure at the core of the absorption band[31], and more complex structures in the wings that could point to contributions from vibrational hot bands[32]. Detailed rotational contour calculations adopting a symmetric top geometry by Kerr et al.[18] showed a good agreement with observations. They conclude that the λ6614 DIB is compatible with large carbon ring molecules with 14-30 C atoms. Marshall et al.[32] use similar molecular parameters and additional hot vibrational bands to produce models that compare very well to the observations of the λ6614 profile towards two sightlines. Ehrenfreund and Foing[33] also explored PAHs as DIB carriers for DIBs with substructure but adopted a slightly different approach. Instead of reproducing the entire band profile shape, they measured the separation of the P- and R- branch peaks to derive the rotational constant B, assuming a rotational excitation temperature of 50 K. They concluded that λ6614 is compatible with PAHs with approximately 40 C atoms. In an alternative scenario, Bernstein et al.[34] reproduced the λ6614 DIB by modelling superposition of two parallel transitions of a molecule with prolate top symmetry. Furthermore, Bernstein et al.[35] performed spectral deconvolution to find that the λ6614DIB can be reproduced as a combination of doublet and a redtail component arising from different carriers.

One of the key difficulties in comparing the DIB band profiles to molecular contour simulations is that there are potentially many combinations of parameters that could reproduce the observed profiles. Kerr et al.[18] for instance listed 4 different combinations of parameters (in the context of

an oblate top) that each could reproduce the λ6614 DIB adequately. Similar issues are at play when one infers molecular properties from the peak separation since that typically depends on both the molecular constants and the rotational excitation temperature, and there is no a priori information that determines either parameter. However, Cami et al.[20] showed that the λ6614 band profiles vary systematically across sightlines: both the P- and R-branch peaks appear to move away from the central Q-branch peak as the band slightly broadens. In other words, the separation between P- and R-branch peaks varies from sightline-to-sightline. They proposed that those variations are due to changes in the rotational temperature from one sightline to another. MacIsaac et al.[22] confirmed this effect using data from the EDIBLES survey[9], and carried out a detailed analysis assuming a linear or spherical molecular geometry. They showed that in that case, the difference between the PQ and QR peak separations is proportional to the rotational constant B and thus allows for direct measurement of B. They found that the λ6614 is compatible with linear molecules of 7-9 C atoms or spherical ones that are 19-27 C atoms in size. The variations from one sightline to another then require changes in the rotational temperature from 13 to 17 K (i.e about 30% difference) as we go from the sightline with the narrowest profile to that with the widest one. It has been proposed that change in rotational temperature causes variations in DIB profiles other than λ6614 as well, such as $\lambda\lambda$5797, 6379[22] and λ4428[36].

In this paper, we investigate the variations in the λ6614 DIB profile in the context of symmetric top molecules, and in particular molecules with an oblate top geometry[37] such as coronene. We will use the detailed modelling work by Kerr et al.[18] and Marshall et al.[32] as a starting point. We will use these simulated profiles to fit the observed profiles of the λ6614. What is new in this paper is that we focus on reproducing the profile *variations* that are observed across different single-cloud sightlines consistently with the same carrier molecule. This will help us investigate the origin of observed variations and constrain the size and geometry of the carrier of λ6614 DIB and the physical conditions of the interstellar environments in which they reside. In Sect. 2, we briefly review the observational data that were analyzed by MacIsaac et al.[22]. In Sect. 3 we present the formalism that we have adopted to simulate the rotational contours of oblate symmetric top

species. Sect. 4 presents a parameter study to explore the impact of individual molecular and environmental parameters on the profile. In Sect. 4.2, we examine the variations in profile shapes across different sightlines and elucidate the environmental parameters potentially contributing to these variations. In Sect. 6, we employ contour modelling for the λ6614 DIB, we fit our model to EDIBLES data aiming to account for variations across sightlines. Using this method, we gain insights into potential molecular and environmental parameters that could reproduce the substructures in λ6614 DIB and also their observed variations. In Sect. 7, we compare our fit results to previous studies, focusing particularly on the influence of environmental parameters that may give rise to variations in the profile. We further discuss the implications in terms of carrier sizes and geometries. We summarize our conclusions in Sect. 8.

2 Observations and targets

Table 1: Basic information about the targets in this work, from MacIsaac et al.[22].

HD	Right ascension	Declination	$v_{\rm ISM}$ [km s^{-1}]	Spectral type	$E(B-V)$ [mag]	$f(H_2)$
23180	03:44:19.1	+32:17:17.7	13.3±0.2	B1 III	0.28	0.51
24398	03:54:07.9	+31:53:01.1	13.8±0.1	B1 Ib	0.29	0.60
144470	16:06.48.4	−20:40:09.1	−10.1±0.1	B1 V	0.21	0.132
147165	16:21:11.3	−25:35:34.1	−6.4±0.3	B1 III	0.37	0.053
147683	16:24:42.7	−34:53:37.5	−0.8±0.2	B3: Vn (SB2)	0.29	0.377
149757	16:37:09.5	−10:34:01.5	−13.9±0.1	O9.2 IVnn	0.32	0.630
166937	18:13:45.8	−21:03:31.8	−6.4±0.2	B8 Iab(e)	0.22	—
170740	18:31:25.7	−10:47:45.0	−10.1±0.2	B2 V	0.45	0.575
184915	19:36:53.5	−07:01:38.9	−12.0±0.2	B0.5 IIIn	0.22	0.366
185418	19:38:27.5	+17:15:26.1	−10.1±0.2	B0.5 V	0.42	0.398
185859	19:40:28.3	+20:38:37.5	−8.2±0.1	B0.5 Ia	0.56	—
203532	21:33:54.6	−82:40:59.1	14.2±0.1	B3 IV	0.30	0.84

The targets and observations selected for this study are exactly the same as those used by MacIsaac et al.[22]. Basic information about these stars is listed in Table 1.

The data originate from the ESO Diffuse Interstellar Bands Large Exploration Study (EDIBLES)[9] that collected spectra of 123 bright O and B type stars over a large spectral range (305–

1042 nm) at high spectral resolution (R~70,000 in the blue and 100,000 in the red arm) and at a high signal-to-noise (S/N~ 1000 per target). The EDIBLES targets were chosen to represent interstellar environments with different physical conditions; consequently, the EDIBLES sample represents a range in interstellar parameters (e.g. interstellar reddening $E(B-V)$, visual extinction A_V, molecular hydrogen fraction f_{H_2}, etc.) along different lines of sight.

For this study, we selected single cloud sightlines, i.e. sightlines that probe one dominant intervening interstellar cloud to avoid complications in the DIB profiles due to contributions from multiple physical environments along a line of sight. MacIsaac et al.[22] presented 12 such sightlines, and we use the same targets in this study; and all observed profiles of λ6614 DIB are rescaled to a depth of 0.9. We refer to MacIsaac et al.[22] for more details on the sightline selection, data processing and stacking and some of the measurements that will be used in this paper.

In addition to the EDIBLES data, we also analyzed the higher resolution data (R~220,000) presented by Galazutdinov et al.[38] and studied in the context of the λ6614 DIB profile by Cami et al.[20].

3 Rotational contours of oblate symmetric tops

3.1 Energy levels

We start by describing the formalism that we used to simulate rotational contours for symmetric top molecules. Much of this formalism is similar to what MacIsaac et al.[22] presented for linear and spherical top species. In that formalism, the observed DIB is the unresolved contour representing all allowed rovibronic transitions from a lower electronic and vibrational state (E, v) to an excited state (E', v') where the prime indicates quantities in the excited state. The lower state will almost always be the ground state when considering interstellar absorption. The molecular geometry determines what rotational states can exist in these lower and upper vibronic states, what transitions are allowed, and the transition probabilities. We furthermore assume that we can characterize the population distribution over the rotational states in the ground state using a single parameter, the

rotational excitation temperature. This then fully determines a set of individual transitions for this band. The spacing between those transitions is small, and intrinsic and instrumental broadening of the lines then results in smooth contours whose possible substructures carry the imprint of this rotational distribution.

In general, a polyatomic molecule can rotate around three different axes, and this defines three rotational constants $A \geq B \geq C$ (typically expressed in either cm^{-1} or MHz) that are inversely proportional to the moments of inertia around each of these axes. For linear and spherical top molecules, only a single rotational constant B is required, and then the energy of a rotational level is described by just one quantum number J, representing the total angular momentum. In this paper however we study symmetric tops, and the rotational energy levels of symmetric tops require two quantum numbers: J and K, where K represents the projection of J onto the molecular symmetry axis. K can take integer values from $-J$ to J, but as we will see below, the energy expressions only include terms in K^2. For an oblate top, $A = B > C$, the rotational energy expression in the ground state is then given (to the first order) by

$$F(J, K) = BJ(J + 1) + (C - B)K^2 \tag{1}$$

where the energies and rotational constants are expressed in wavenumbers (cm^{-1}). For $K = 0$, this expression reduces to that for the rotational energy of linear or spherical top species. For $K \neq 0$, the second term in the expression is negative for oblate species since $B > C$, and thus the energies are lowered.

For species like coronene, the origin (i.e. vibrationless) transition to the excited electronic state is forbidden. However, if $v' \neq v$, then Coriolis-induced vibrational transitions can be possible. This then requires inclusion of a Coriolis distortion term in the energy expression for the rotational levels in the excited state:

$$F'(J', K') = B'J'(J' + 1) + (C' - B')K'^2 \pm 2C'\zeta'K' + C'^2 \tag{2}$$

where ζ' is the Coriolis coupling constant (in cm^{-1}). The Coriolis distortion term (the third term, the one containing ζ') in that equation is positive for transitions with $\Delta K \equiv K' - K = +1$[1] and negative for ΔK = -1. Note that the rotational constants in the excited state are normally close to those in the ground state. Consequently, it is common to specify the relative change in rotational constant, e.g. $\Delta B/B = (B' - B)/B$ rather than the upper state rotational constants $B'$and C'. For what follows, we will thus often express ΔB as a percentage; what is meant is then of course technically $100 \times(\Delta B/B)$.

3.2 Transition frequencies

Selection rules on J and K specify the possible transitions between rotational energy levels from the ground state to those in the excited state. Just like for linear and spherical top species, the selection rules on J are:

$$\Delta J = 0, \pm 1 \tag{3}$$

and this results in three sets of spectral lines with ΔJ = -1, 0, and +1 respectively, and these are called the P, Q and R branches respectively. In addition, symmetric tops have selection rules on K that depend on the type of transition. For such molecules, the change in dipole moment that happens in a transition can be parallel or perpendicular to the principal molecular axis, and consequently, they are called parallel and perpendicular transitions. Both types of transitions have different selection rules on K:

$\Delta K = 0$ for parallel transitions

$\Delta K = \pm 1$ for perpendicular transitions

Thus, perpendicular bands have more allowed individual transitions than parallel bands. For the remainder of this paper, we focus on perpendicular transitions, consistent with the transitions from the ground state of coronene. For perpendicular bands, we then find that each of the three branches (P, Q and R) will exhibit sub-bands with either ΔK = -1 or +1 which are then denoted

[1]For the remainder of this paper, any difference Δ implies a difference between an upper state quantity and the corresponding lower state one.

by lowercase prefix p and r, respectively. Thus, for a symmetric top, there are 6 branches for transitions between rotational energy levels and these are labelled pP, rP, pQ, rQ, pR and rR.

Once the rotational levels have been calculated using Eqs. (1) and (2), and the appropriate selection rules have been applied, it is then straightforward to calculate the frequency of the transition (in wavenumbers):

$$\nu = \nu_0 + F'(J', K') - F(J, K) \tag{4}$$

where $\nu_0 = E' + v' - E - v$ is the frequency of the band origin.

3.3 Intensities

For each of the transitions obtained from Eq. (4), we then need to calculate the absorption intensity I_ν. That transition originates from a rotational level specified by J and K, and we will denote the population of that level with n_{KJ}. The intensity of a transition out of that level to an upper level is then:

$$I_\nu \propto A_{KJ} n_{KJ} \tag{5}$$

where A_{KJ} is the Hönl-London factor, which is proportional to the square of transition moment summed over all orientations of J. Convenient expressions for A_{KJ} for transitions in each of the branches are taken from Herzberg [39]:

$$A_{KJ} = \frac{(J + 2 \pm K)(J + 1 \pm K)}{(J + 1)(2J + 1)} \qquad (\text{for } \Delta\mathrm{J} = +1) \tag{6}$$

$$A_{KJ} = \frac{(J + 1 \pm K)(J \mp K)}{J(J + 1)} \qquad (\text{for } \Delta\mathrm{J} = 0) \tag{7}$$

$$A_{KJ} = \frac{(J - 1 \mp K)(J \mp K)}{J(2J + 1)} \qquad (\text{for } \Delta\mathrm{J} = -1) \tag{8}$$

where the upper signs refer to $\Delta K = +1$ and the lower signs to $\Delta K = -1$.

For the second factor in Eq. (5), the population n_{KJ}, we will assume here that we can describe the rotational population distribution in the ground state by a single parameter, the rotational tem-

perature $T_{\rm rot}$ and that the population follows a thermal distribution. In that case, we can use the Boltzmann equation and write that

$$n_{KJ} \propto g_J \exp(-\frac{hcF(J,K)}{kT_{\rm rot}}) \tag{9}$$

where $g_J = (2J + 1)$ is the statistical weight of level J; $F(J, K)$ is the rotational energy (in cm^{-1}) from Eq. (1), $T_{\rm rot}$ is the rotational temperature in Kelvin, and h, c and k are respectively Planck's constant, the speed of light and Boltzmann's constant all in cgs units.

Eqs. (4) and (5) provide us with a complete line list containing frequencies and intensities for all allowed transitions. Typically, one then convolves these transitions with an intrinsic line profile that represents physical line broadening mechanisms, then carries out radiative transfer, and finally convolves the resulting intensity with an instrumental broadening profile. This final step removes the signatures of the individual transitions and instead produces a smooth band profile. To simplify this operation further, we assume here that both the intrinsic and instrumental profiles are Gaussian (but with different widths), and that the absorption is optically thin. In that case, the depth of the intensity scales linearly with the optical depth, and we can combine the intrinsic and instrumental convolution into a single operation – a convolution with a Gaussian of width σ which then represents a combination of intrinsic and instrumental widths. At a typical spectral resolving power of 75,000, an unresolved line would have a full width at half maximum (FWHM) of 4 km s^{-1} or 0.2 cm^{-1} for the λ6614 DIB; this corresponds to $\sigma = 0.085$ cm^{-1}. The final step in our calculation is to scale the profiles to a maximum depth of 10% of the continuum, comparable to the observed depth of the λ6614 DIB profile.

3.4 Computational details and benchmarking

We implemented the formalism described above using a series of Python codes. We first calculate the rotational energies $F(J, K)$ and $F'(J', K')$ up to $J = 1000$. For the range in rotational constants and temperatures, we consider in this paper, higher-J terms provide negligible contributions to the

profile. We then create a line list by applying the appropriate selection rules and calculate the line intensities using Eq. (5). Finally, we adopted a Gaussian smoothing kernel with width σ (expressed in cm^{-1}) that combines intrinsic and instrumental effects as discussed above.

To debug and benchmark our Python codes, we used PGOPHER, a widely used general-purpose program for simulating and fitting rotational, vibrational and electronic spectra[40]. Since PGOPHER doesn't support the fitting procedures used in this study, we had to develop Python codes to simulate spectra instead of using PGOPHER itself. We compared both our line lists and the resulting smoothed line profiles with the ones that PGOPHER produces, adopting the input parameters listed in Table 2; note that these are taken from Kerr et al. [18], and more specifically their condition c (see also Table 3).

Table 2: Input parameters used for benchmarking our codes and comparing them to the output of PGOPGHER.

Parameter	Value
T_{rot}	61.2 K
Point Group	D6h
Ground state configuration:	
B	0.00336 cm^{-1}
C	0.00168 cm^{-1}
Symmetry	A1g
Excited state configuration:	
Band origin (ν_0)	0 cm^{-1}
B'	0.003354288 ($\Delta B = -0.17\%$)
C'	0.001677144 ($\Delta C = -0.17\%$)
ζ'	-0.49
Symmetry	E1u
σ	0.1953 cm^{-1}

We find that the PGOPHER's resulting profile as well as our calculated line profile completely

overlap; the largest residuals are ~0.04%. We found the same level of comparison to hold for simulations using other sets of parameters

4 A parameter study

As elaborated in the prior section, our model incorporates five key parameters that will affect the precise shape of the rotational contours: three tied to the molecular size and geometry (B, ΔB, ζ) and two to environmental and instrumental factors ($T_{\rm rot}$ and σ). In this section, we investigate the influence of each of the parameters on the resulting profile and describe what types of band profiles we can obtain for what combination of parameters.

Before presenting our detailed models, we summarize here the assumptions, constraints and limitations of this study. We only consider perpendicular transitions of planar oblate tops (as these transitions have shown a good agreement with the observations, see Sect. 1), i.e. molecules with a disk-like appearance, such as members of the coronene family. Such species have $A = B > C$ and more specifically $B = 2C$. We assume that this geometry is preserved in the excited state, i.e. $\Delta B/B = \Delta C/C$. We assume that the rotational population follows a Boltzmann distribution characterized by the rotational temperature $T_{\rm rot}$ and that the absorption is optically thin.

4.1 Defining the parameter space

We next need to define reasonable ranges for each of the parameters. The first range to consider concerns the rotational constant B, which is inversely proportional to the moment of inertia of the molecule and thus its size. If the DIB carriers are PAH-like planar molecules, their size should be at least about 20 carbon atoms since small PAHs are thought to be unstable against photo-dissociation in the ISM[41]. That is about the size of coronene ($C_{24}H_{12}$), the smallest PAH with an oblate top geometry. Coronene has $B = 1.11 \times 10^{-2}$ cm^{-1} [42], and we thus chose as upper limit a slightly higher value of $B = 5 \times 10^{-2}$ cm^{-1} corresponding to slightly smaller species. At the upper end of the size range, we consider species smaller than 100 atoms, and then a good reference species

is circumcircumcoronene ($C_{96}H_{24}$) with a rotational constant of $B = 7.0 \times 10^{-4}$ cm^{-1} [43]. We thus considered B-values in the range $5 \times 10^{-2} - 5 \times 10^{-4}$ cm^{-1}. For our parameter study, we selected five logarithmically equidistant values between these two extremes.

For large molecules like the members of the coronene family, geometric distortions upon excitation are fairly small. Moreover, the atoms in vibrationally excited molecules will – on average – spend more time further away from the main molecular axis. Thus, the moment of inertia will be slightly larger in the excited state and consequently the rotational constant slightly smaller. For our parameter study, we consequently considered values for $\Delta B/B$ that are small and negative. Again we picked five values: 0.0, -0.2, -0.4, -0.6 and -0.8%. This corresponds well to the range considered by Kerr et al. [18]. The final molecular parameter to consider is then the Coriolis coupling term, ζ, which can range between -1 and +1 cm^{-1} and we picked the following five values: -1, -0.5, 0, 0.5 and 1 cm^{-1}.

Finally, we have two environmental parameters to consider: the rotational excitation temperature $T_{\rm rot}$ and the linewidth σ. Without any prior knowledge of the excitation and cooling of these species, it is difficult to firmly establish what the temperature should be. Polar molecules can cool efficiently and could be in thermal equilibrium with the cosmic microwave background radiation. Thus, the lowest value we consider is 2.7 K. However, UV excitation and fluorescent IR cooling of large PAHs causes the molecules to spin up [44], resulting in equilibrium temperatures of ~100 K. Again, we choose 5 values to sample this range and thus used $T_{\rm rot} = 2.7, 10, 30, 70$ and 100 K.

The parameter space for linewidth (σ) explored here ranges from 0.05cm^{-1} (~1 km s^{-1}) to 0.3 cm^{-1}(~6 km s^{-1}). That corresponds well to the typical range of line widths seen in interstellar clouds; for example, Crane et al. [45] determined b values in the range of 1–4 km s^{-1} for interstellar CH and CH$^+$ lines towards μ Sgr. The five selected values within this range are 0.05, 0.11, 0.17, 0.23, 0.3 cm^{-1}.

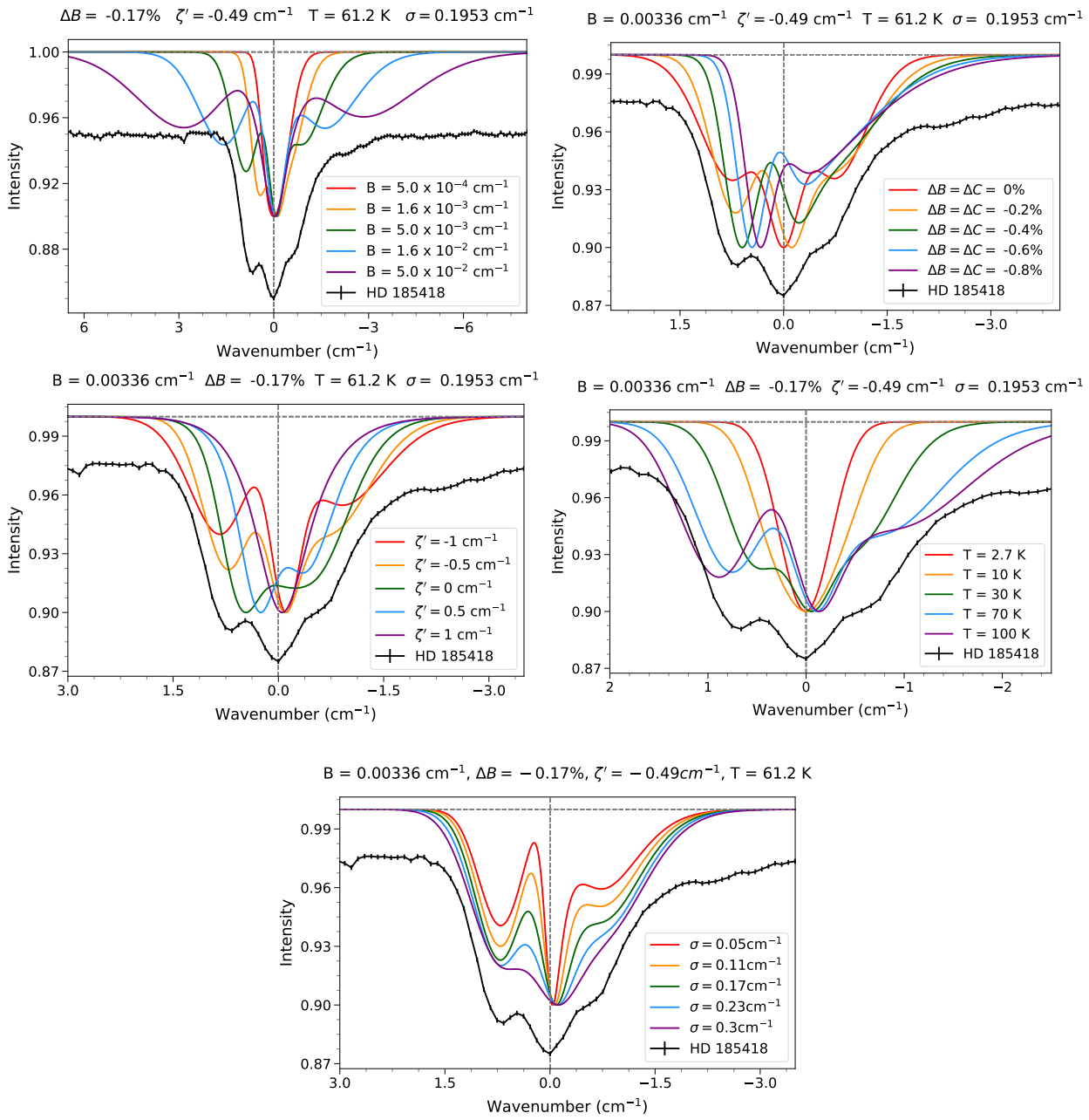


Figure 1: Line profiles for our reference model for 5 different values of (a) B, (b) ΔB, (c)ζ , (d) T_{rot}, (e) σ.

4.2 The influence of each parameter on the profile

We now investigate the influence of each of the five parameters ($B, \Delta B, \zeta, T_{\rm rot}, \sigma$) on the resulting line profile. Our reference point is the model that corresponds to the parameters listed in Table 2 As was mentioned before, this is one of the models by Kerr et al.[18] that can reproduce the profile of the λ6614 DIB. We will now change the parameters one by one, and describe the influence on the band profile.

We first investigate the effect of B. Fig. 1 (a) shows our reference model but with different values of B. The lowest values of B result in a single narrow absorption band, and an increase in B broadens the profile. Asymmetries in the profile are noticeable at all values of B in our chosen range. The R-branch is more pronounced than the P-branch. The R-branch peak separates from the Q-branch peak at a lower value of around B = 0.001 cm^{-1}, but the P-branch peak only above the B = 0.005 cm^{-1}. As B increases, the peak position of the Q branch subtly shifts toward lower wavenumbers, though the effect is quite minimal.

The main influence of the next parameter, ΔB, is to make the profiles more asymmetric. This can profoundly change the resulting profile, and affects the number of peaks that are noticeable in the profile (See Fig. 1 (b)). At ΔB = 0%, the profile is the most symmetric, and the 3 P-, Q- and R-subpeaks are clearly distinguishable, with the P- and R-branch peaks similar in depth. As soon as we introduce even a slight distortion (ΔB=-0.2%), the profile becomes quite asymmetric. The R-branch becomes more peaked and moves closer to the band origin, while the Q- and P-branches broaden and move away from the band origin, and eventually merge into a single broad red wing. When ΔB reaches -0.4%, the band now looks more like a double-peaked profile. For the lowest ΔB value of -0.8%, even the Q-branch is not very pronounced anymore.

The next molecular parameter we consider is ζ. The impact of ζ on the band profile is illustrated in Fig. 1 (c). It mainly affects the separation of the P and R branches from the Q branch. As ζ increases from -1 to 1, the P and R branches move closer to the Q branch. Consequently, we see a clear triple-peak structure at ζ = -1 cm^{-1}; At ζ = 0 cm^{-1}, the profile exhibits a double peak structure and finally just a single peak at ζ=+1 cm^{-1}.

Fig. 1 (d) shows models for the same parameters as our reference model, but for the 5 different temperatures that we consider here. At the lowest temperatures, there is only a single, narrow absorption band. As the temperature increases, the profile broadens and at $T_{\rm rot} \approx 30$ K, the triple-peak substructure starts to appear. At higher temperatures still, the profile broadens more, and the sub-peaks move away from each other. It should be noted here that the peak position of the Q branch shifts to a slightly lower frequency with increasing temperature. This occurs because $\Delta B \neq 0$, and consequently the Q-branch lines do not all have the same frequency. Such an effect also occurs for linear and spherical top species; see e.g. Eq. (A.15) in MacIsaac et al.[22]. Note that in all models, the R-branch sub-peak (at positive wavenumbers) is slightly more pronounced than the P-branch peak.

Finally, Fig. 1 (e) illustrates the role of the intrinsic linewidth. All branches are broadened and the peaks become less pronounced as one would expect, but we also want to point out that the precise position of the peaks changes slightly with σ. This is perhaps best seen in the Q-branch that shifts slightly to the red (lower frequencies) for larger line widths; similarly, the R-branch shifts slightly to the blue (larger frequencies). These shifts are the consequences of asymmetries in the profiles. Comparing Fig. 1 (d) and Fig. 1 (e)indicates that profiles at lower ζ and higher $T_{\rm rot}$ resemble each other, and same is the case of higher ζ and lower $T_{\rm rot}$. Thus, high-resolution data is required to distinguish between them.

Table 3: Parameters derived from fitting simulated contours to the λ6614 diffuse interstellar band. Taken from Kerr et al.[18]

	T_{rot} (K)	B' cm^{-1}	ΔB (%)	Γ cm^{-1}	ζ	χ^2
a	8.9	0.01913	-0.85	0.32	-0.46	2.12
b	20.2	0.00947	-0.42	0.37	-0.43	1.79
c	61.2	0.00336	-0.17	0.46	-0.49	1.57
d	101.3	0.00286	-0.21	0.47	-0.54	2.07

5 What causes variations in the λ6614 DIB profiles?

We extended our parameter study from the previous section and investigated various combinations of parameters that can produce a profile similar to λ6614 DIB. We find that, as pointed out by Kerr et al.[18], there are several combinations of parameters that can reproduce the λ6614 DIB profile in a sightline. We next consider the extra information we have for this DIB, and the new aspect of this paper: that the line profile shows (subtle) changes in different sightlines, and that in particular the peak separation varies from one sightline to another. Since the molecular parameters for the carrier ought to remain the same in each sightline, Cami et al.[20] and MacIsaac et al.[22] suggest that the only way we can explain such variations is by changes in the rotational excitation temperature $T_{\rm rot}$. However, as we will show below, the line width σ also contributes to the observed profile variations, alongside $T_{\rm rot}$.

Fig. 2 shows two EDIBLES observations of the λ6614 DIB: the profile for HD 185418 (red curve; representative for the smallest peak separation in our sample) and for HD 166937 (green; representative for the largest peak separation). The P and R peak positions for both objects are clearly different compared to the Q branch peak. We first investigate whether the variations in the observed profile can be reproduced by only changing the temperature.

We again start from model c presented by Kerr et al.[18]. Since the R-branch is most pronounced and the peak position best determined, we first changed the temperature in our models until the R-branch peak position of the model corresponded to the minimum position indicated by the dotted line; this was the case for $T_{\rm rot}$ = 40 K. We then increased the temperature until the R-branch peak position matched the other extreme, and this is the case for $T_{\rm rot}$ = 55 K.Those models are shown in the left panel of Fig. 2.

The models readily demonstrate that the observed variations in peak separation can indeed be attributed to temperature changes. However, the models also show that changing the temperature should affect the position of the red and blue wings relative to the Q-branch peak as well, and this effect is not seen in the observations. Moreover, the Q-branch peak itself appears markedly broader in the observation with the largest peak separation, but that broadening is not obvious in

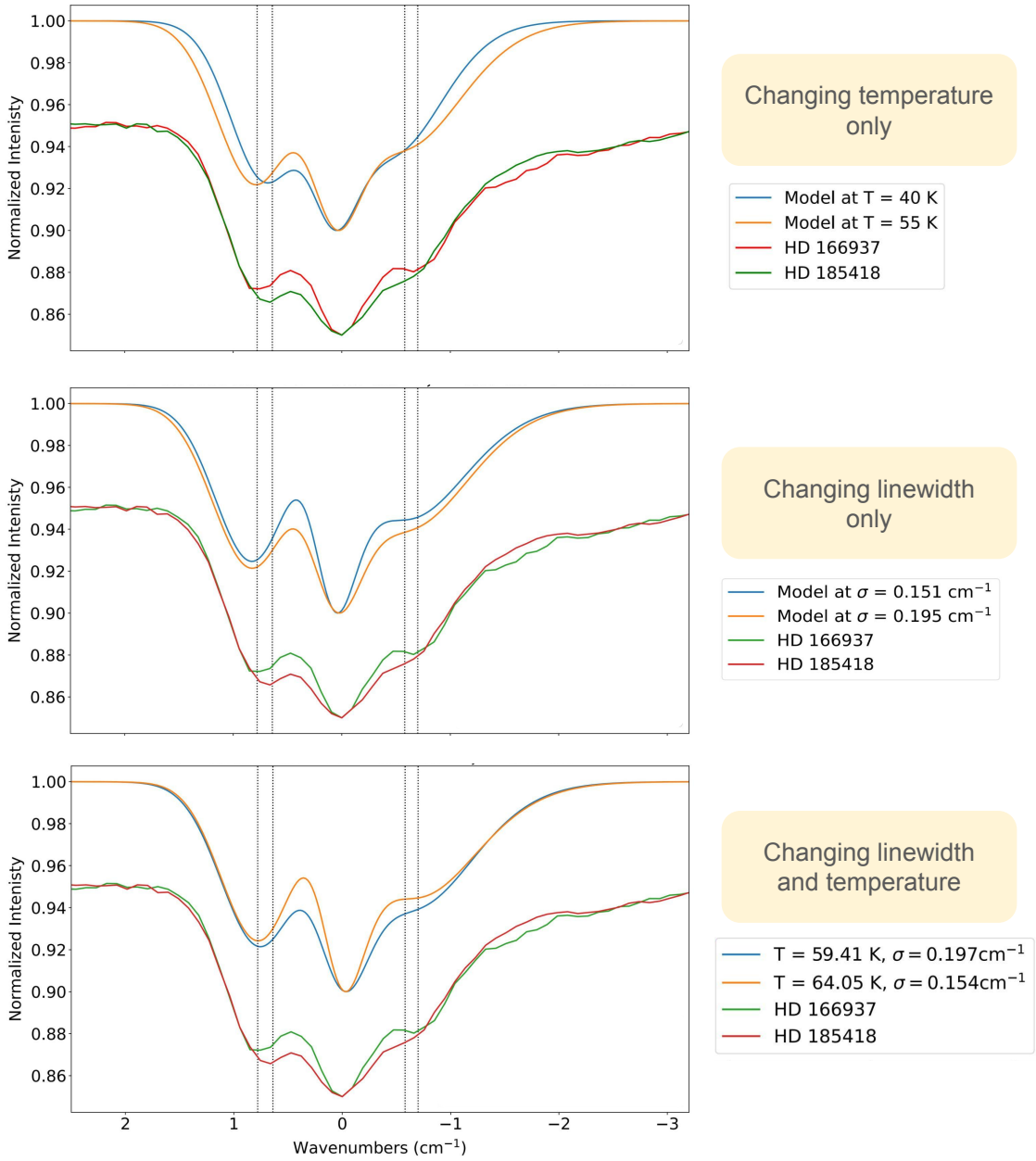


Figure 2: A comparison of simulated band profiles for the λ6614 DIB with observed profiles, illustrating the need to consider both temperature and line width as free parameters in the fitting. In all panels, the red spectrum is the EDIBLES spectrum for HD 185418, the target with the smallest measured peak separation. The green spectrum is that of HD 166937 which has the largest peak separation in our sample. Both profiles are centered on the Q-branch peak. Vertical dotted lines in all panels indicate the extreme positions of the P- and R-branch peaks relative to the Q-branch peak, as measured by MacIsaac et al.[22]. The blue and yellow curves are various simulated profiles, all starting from the parameters of model c (see Table 3), and shifted so that they are centered on the Q-branch peak. *(Top)* Models with different temperatures, but using a fixed line width; *(Middle)* Models with different line widths, but using a fixed temperature; *(Bottom)* Models where temperature and line width are changed at the same time.

the models, and can thus not be attributed to changes in the temperature.

These considerations led us to explore the role of the line width σ in the profile variations. Using model c from Kerr et al.[18] once more as our benchmark, we adjusted σ to reproduce the changing width of the Q-branch peak (see the right panel in Fig. 2). As expected, an increase in σ leads to a broader Q-branch peak, and it additionally also reduces the peak-to-peak contrast in the profile, which is also the case in the observations. While the wings of the two models do not precisely align, they closely resemble each other. The peak positions of the P- and R-branches do not show much difference between these two models, but given the asymmetric nature of the profiles, line width variations may affect the precise peak position a bit. The possible role of the line width becomes even more clear when we include the line width as a parameter in our fitting. This significantly improves the quality of the fit, and reproduces the Q-branch width, peak separation and red and blue wings at the same time (see the bottom panel in Fig. 2). Note that allowing σ to vary, results in higher $T_{\rm rot}$.

Summarizing, we find that changes in the rotational temperature can account for differences in peak separation, yet they do not reproduce other observed changes to the profile such as the width of the Q-branch peak and the precise position of the blue and red wings. These latter variations suggest that we also need to consider variations in σ when comparing different sight lines. The overall profile variations from one sightline to another thus likely stem from combined changes in both the rotational temperature and the line width.

6 Contour modelling

We now fit the model described in Sect.3 to the EDIBLES observations (see Sect.2). We used the `lmfit`[46] Python package adopting a Levenberg-Marquardt algorithm. While the model has five primary parameters that determine the exact profile shape, we also allowed for small adjustments in the band origin frequency during fitting to account for the wavelength calibration issues and uncertainties in radial velocity correction and to ensure the best possible alignment between models and observations (See Sect. 2 and refer to[22] for more details). In the fitting process, parameter

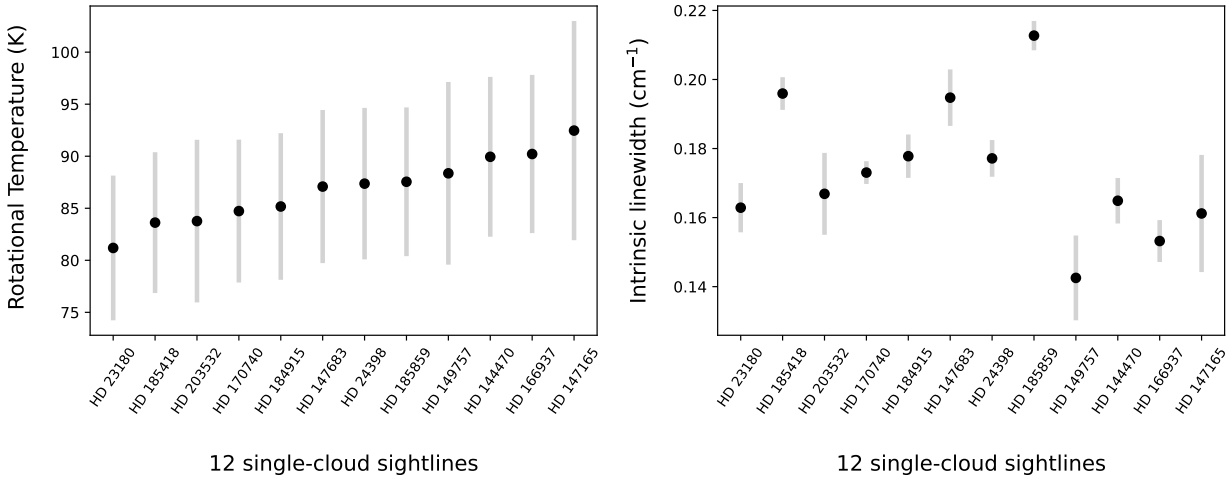


Figure 3: Variations in rotational temperature and line width corresponding to the best-fit models for the λ6614 DIB profile. The sight lines are shown in order of increasing rotational temperatures in both panels.

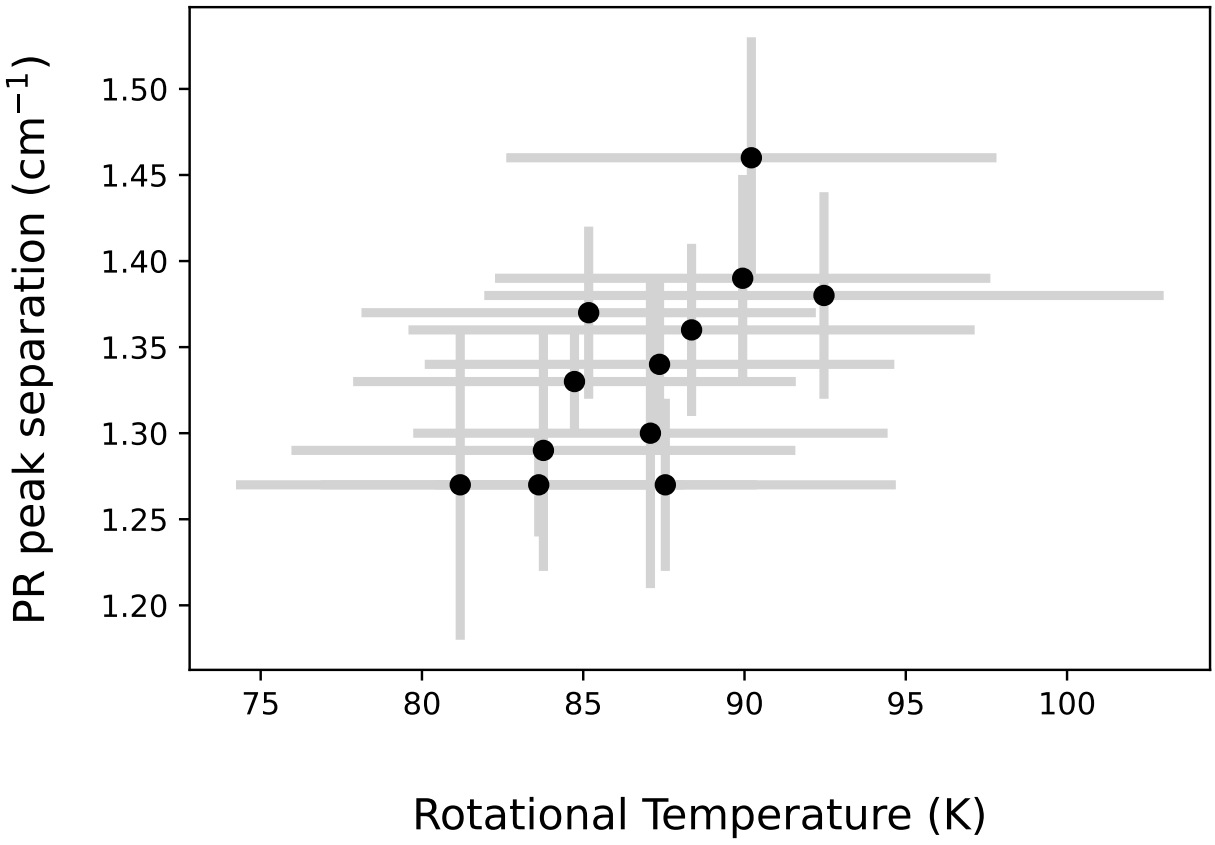


Figure 4: T_{rot} as a function of PR peak separation as measured by MacIsaac et al.[22]

constraints were kept in the ranges described in Sect. 4 (although we did occasionally extend the range for T_{rot} to 500K to avoid the best fit parameters being pegged at the boundary of the parameter space). For each of the fitting exercises described below, we ran the fitting several times, each time starting from different initial conditions. This allows us to verify that our best-fit solutions do not correspond to a local minimum, and are thus more likely to correspond to the global minimum.

6.1 Fitting each sightline independently

We first fitted a model to each of our sightlines individually, allowing all parameters to vary. Our primary goal is to reproduce the detailed profile substructures rather than the entire profile. However, when we included the entire band in the fitting process, we found that much weight was given to the red wing, resulting in poor fits for the peaks. This red wing may be due to contamination by a hot band[32] and overlaps with the main feature[35]. We therefore chose to not include the red wing. Instead, we used all data points whose (rescaled) absorption depth is 5% or more (i.e. their normalized fluxes are lower than 0.95), as well as the blue wing (positive wavenumbers) in the fitting. This corresponds to the criterion for fitting that was also used by Bernstein et al.[35]. These individual fits are shown in blue in Fig. 5.

In general, the best-fit models reproduce the sub-peak structure well, and often also the blue wing, but they do not reproduce the red wing at all. The best fit parameters are listed in Table 4. We find that the value of B ranges from $(7.0 \pm 1.0) \times 10^{-4}$ to $(3.7 \pm 2.5) \times 10^{-3}$ cm^{-1} and T_{rot} ranges from (52 ± 30) to (370 ± 100) K. Other parameters correspondingly also have a wide range. Our fitting confirms that – without any additional constraints – the triple peak structure can be reproduced by many different combinations of parameters. Hence, we need to fit all sightlines at once to constrain the combination of parameters that can reproduce λ6614 DIB.

6.2 Fitting all sightlines at once

To now account for the variations in the profiles from one sightline to another, we conducted a simultaneous fit for all 12 sightlines in our study. In this exercise, all parameters are allowed to

Table 4: Best-fit parameters when fitting each sightline independently (using the EDIBLES data).

HD	B	ΔB	ζ	T	σ	origin shift
	[$\times 10^{-3}$ cm^{-1}]	[%]	[cm^{-1}]	[K]	[cm^{-1}]	[cm^{-1}]
23180	2.6 (± 1.1)	-0.08 ± 0.02	-0.35 ± 0.04	66 ± 30	0.19 ± 0.02	0.00 ± 0.01
24398	2.9 (± 0.9)	-0.06 ± 0.02	-0.27 ± 0.02	68 ± 20	0.17 ± 0.01	-0.04 ± 0.01
144470	7.0 (± 1.0)	-0.01 ± 0.00	-0.41 ± 0.06	370 ± 100	0.23 ± 0.02	-0.02 ± 0.01
147165	1.8 (± 3.4)	0.00 ± 0.02	-0.31 ± 0.04	$110\ ^{+200}_{-110}$	0.17 ± 0.02	-0.07 ± 0.03
147683	3.7 (± 2.5)	-0.04 ± 0.03	-0.30 ± 0.03	52 ± 30	0.20 ± 0.01	-0.02 ± 0.01
149757	2.0 (± 1.2)	-0.06 ± 0.03	-0.44 ± 0.04	89 ± 50	0.20 ± 0.01	-0.02 ± 0.01
166937	1.9 (± 0.5)	-0.06 ± 0.01	-0.34 ± 0.03	102 ± 30	0.17 ± 0.01	0.05 ± 0.01
170740	2.4 (± 0.4)	-0.08 ± 0.01	-0.30 ± 0.02	78 ± 10	0.18 ± 0.01	0.01 ± 0.01
184915	7.0 (± 1.0)	-0.03 ± 0.00	-0.31 ± 0.03	320 ± 70	0.20 ± 0.02	0.13 ± 0.01
184518	3.0 (± 0.3)	-0.10 ± 0.01	-0.29 ± 0.01	64 ± 5	0.20 ± 0.01	0.06 ± 0.00
185859	2.3 (± 0.4)	-0.08 ± 0.01	-0.24 ± 0.01	90 ± 20	0.19 ± 0.01	0.03 ± 0.00
203532	2.6 (± 0.4)	-0.10 ± 0.01	-0.33 ± 0.02	70 ± 10	0.18 ± 0.01	0.07 ± 0.01

Table 5: Best-fit parameters for the 6614 DIB profile for both the EDIBLES data set and the data presented by Galazutdinov et al.[38], when fitting all sightlines at once. Velocity dispersions (*b*-values) for the KI and CH^+ lines in the EDIBLES data.

λ6614 DIB											KI	CH^+
EDIBLES data						Galazutdinov data						
Molecular parameters												
B		ΔB		ζ		B		ΔB		ζ		
[$\times 10^{-3}$ cm^{-1}]		[$\times 10^{-2}$ %]		[cm^{-1}]		[$\times 10^{-3}$ cm^{-1}]		[$\times 10^{-2}$ %]		[cm^{-1}]		
2.2±1.8		-7.2 ± 0.4		2.9 ± 0.1		2.4±0.3		-4.3 ± 0.4		3.0 ± 0.1		
Environmental parameters												
HD	T_{rot}	σ		σ	origin shift	T_{rot}	σ		origin shift		b	b
	(K)	[$\times 10^{-1}$ cm^{-1}]		(km s^{-1})	[$\times 10^{-1}$ cm^{-1}]	(K)	[$\times 10^{-1}$ cm^{-1}]		[$\times 10^{-1}$ cm^{-1}]		(km s^{-1})	(km s^{-1})
23180	81 ± 7	1.62 ± 0.07		3.2 ± 0.1	0.11 ± 0.09						1.7 ± 0.2	2.8 ± 0.1
24398	87 ± 7	1.77 ± 0.05		3.5 ± 0.1	-0.22 ± 0.07						0.7 ± 1.2	2.4 ± 0.2
144217						87 ± 10	1.60 ± 0.07		0.24 ± 0.10			
145502						83 ± 10	1.47 ± 0.05		-0.24 ± 0.07			
144470	90 ± 8	1.64 ± 0.06		3.3 ± 0.1	-0.04 ± 0.10	87 ± 10	1.54 ± 0.06		-0.82 ± 0.09		2.9 ± 0.4	2.4 ± 0.1
147165	92 ± 10	1.61 ± 0.16		3.2 ± 0.3	-0.26 ± 0.30	87 ± 10	1.57 ± 0.06		-0.59 ± 0.07		0.3 ± 2.1	2.1 ± 0.1
147683	87 ± 7	1.94 ± 0.08		3.9 ± 0.2	0.11 ± 0.11						4.4 ± 0.4	5.6 ± 0.3
149757	88 ± 9	1.42 ± 0.12		2.8 ± 0.1	-0.03 ± 0.20	89 ± 10	1.64 ± 0.07		0.57 ± 0.10		0.6 ± 1.6	2.4 ± 0.1
166937	90 ± 8	1.53 ± 0.06		3.0 ± 0.1	0.55 ± 0.08						0.4 ± 1.3	2.3 ± 0.1
170740	84 ± 7	1.73 ± 0.03		3.4 ± 0.1	0.10 ± 0.03						2.0 ± 0.1	3.7 ± 0.07
179406						77 ± 10	1.69 ± 0.05		0.59 ± 0.06			
184915	85 ± 7	1.77 ± 0.06		3.5 ± 0.1	0.98 ± 0.08	83 ± 10	1.79 ± 0.06		0.02 ± 0.08		0.3 ± 0.9	2.8 ± 0.1
185418	83 ± 7	1.95 ± 0.04		3.9 ± 0.1	0.64 ± 0.05						2.4 ± 0.2	4.9 ± 0.2
185859	87 ± 7	2.12 ± 0.04		4.2 ± 0.1	0.18 ± 0.04						0.7 ± 0.1	3.3 ± 0.1
203532	83 ± 8	1.66 ± 0.11		3.3 ± 0.2	0.66 ± 0.17						1.4 ± 0.1	2.9 ± 0.1

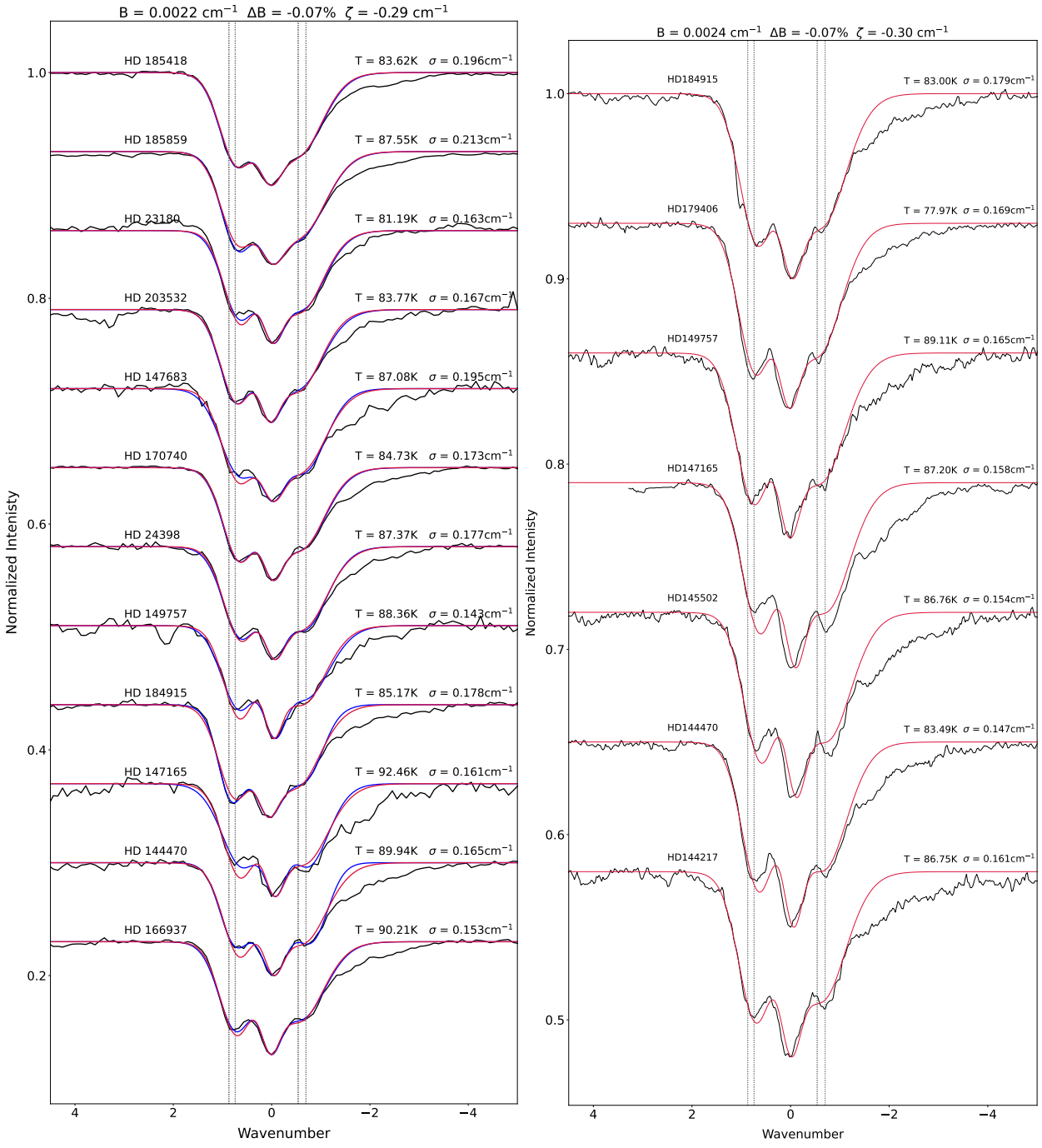


Figure 5: Best-fit models compared to the EDIBLES observations (left panel) and to the higher resolution data presented by Galazutdinov et al.[38] (right panel). Red curves show the best-fit models for which the molecular parameters were held constant across all sightlines; blue curves show models where all parameters are held free for each sightline.

vary in different iterations, but for each iteration, we force all molecular parameters ($B, \Delta B, \zeta$) to be the same across all sightlines, while we allow the environmental parameters ($T_{\rm rot}, \sigma$) to change for each sightline. The resulting fits are shown in Fig. 5 (a). The quality of the fits is not as good as in the individual fits, but is overall aceptable (when ignoring the red wing). We find that the best fits correspond to $B = (2.2 \pm 1.8) \times 10^{-3}$ cm^{-1}, $\Delta B = (-7.2 \pm 0.4) \times 10^{-2}$%, and $\zeta = (2.9 \pm 0.1) \times 10^{-1}$ cm^{-1}; rotational temperatures range from (81 ± 7) to (92 ± 10) K and line widths from $(1.42 \pm 0.12) \times 10^{-1}$ cm^{-1} to $(2.12 \pm 0.04) \times 10^{-1}$ cm^{-1}. The resulting parameters now cover a much narrower range.

Our results show significant sightline-to-sightline differences in the line width. This indicates that the line width is not only a necessary parameter to include as we deduced above, but it also suggests that the line width may play a key role in the variations we observe in the λ6614 profiles. To further investigate what may drive the profile variations, we show in Fig. 3 the variations in $T_{\rm rot}$ and σ and their nominal uncertainties across our sightlines. These figures show that there are indeed small variations in the rotational temperature across our sightlines. The difference between the highest and lowest temperature is however comparable to the nominal uncertainties on the temperature. The variations in the line width are more significant, suggesting that for reproducing the profile variations, this parameter is perhaps the more important one. The line width appears to be independent of the rotational temperature (the correlation coefficient between these two parameters is -0.25). We also find that the rotational temperature certainly influences the peak separation as measured by MacIsaac et al.[22]; the two quantities correlate well (with a correlation coefficient of 0.71; see Fig. 4). Interestingly, we find that the peak separation shows a negative correlation with the line width σ (with a correlation coefficient of -0.61). The peak separation is thus also significantly affected by the line width.

To further corroborate our results, we also fitted our model to the higher resolution data presented by Galazutdinov et al.[38] and studied by Cami et al.[20] in the context of the λ6614 profile variations. These observations are at a resolving power of ~220,000, corresponding to an instrumental width of $\sigma = 0.03$ cm^{-1} or a velocity width of ~1.4 km s^{-1} (FWHM). Note however that

this data set does not correspond to the exact same targets as the EDIBLES data. Those data and corresponding best-fit models are shown in Fig. 5 (b). We find overall similar results: the substructure and variations can be reproduced with $B = (2.4 \pm 0.3) \times 10^{-3}$ cm^{-1}, $\Delta B = (-4.3 \pm 0.4) \times 10^{-2}$%, and $\zeta = (3.0 \pm 0.1) \times 10^{-1}$ cm^{-1} with temperatures ranging from (77 ± 10) to (89 ± 10) K. Our line widths range from (1.47 ± 0.05) × 10^{-1} cm^{-1} to (1.79 ± 0.06) × 10^{-1} cm^{-1}, much larger than the instrumental resolution of 0.03 cm^{-1}, and similar to the range in line widths in the EDIBLES data, even though the resolving power is much higher. This suggests that the line width parameter in our models really traces the intrinsic line width, and is not much affected by the instrumental spectral resolution. For these data, the correlation between $T_{\rm rot}$ and peak separation is 0.44. However, we now find an even stronger anti-correlation of -0.88 between σ and the peak separation.

7 Discussion

For both the EDIBLES data set and the observations at much higher spectral resolution presented by Galazutdinov et al.[38] and studied by Cami et al.[20], we find that the λ6614 DIB (for all the targeted single cloud sightlines) is best reproduced by a species with a rotational constant of B = (2.4 ± 0.3)×10^{-3} cm^{-1}. This is significantly smaller than what Cami et al.[20] and MacIsaac et al.[22] found which was derived based exclusively on the peak separation and not contour simulations. Moreover, they assume that the potential carrier is a linear or spherical species. For the oblate top symmetry used in our simulations, the derived rotational constant corresponds well to a PAH-like molecule with ~54 C atoms such as circumcoronene $C_{54}H_{18}$; $B = 2.2 \times 10^{-3}$ cm^{-1}; [42]. The derived rotational temperatures correspond well to what is expected for such species in the interstellar medium see e.g.[41,44].

We note however that our simulations are not molecule-specific, and furthermore that there are significant discrepancies between our models and observations especially in the red wing of the λ6614 DIB profile. Including centrifugal distortion reduces the discrepancies in the red wing; however, this then makes the fit to the substructures much worse. The red wing could be due

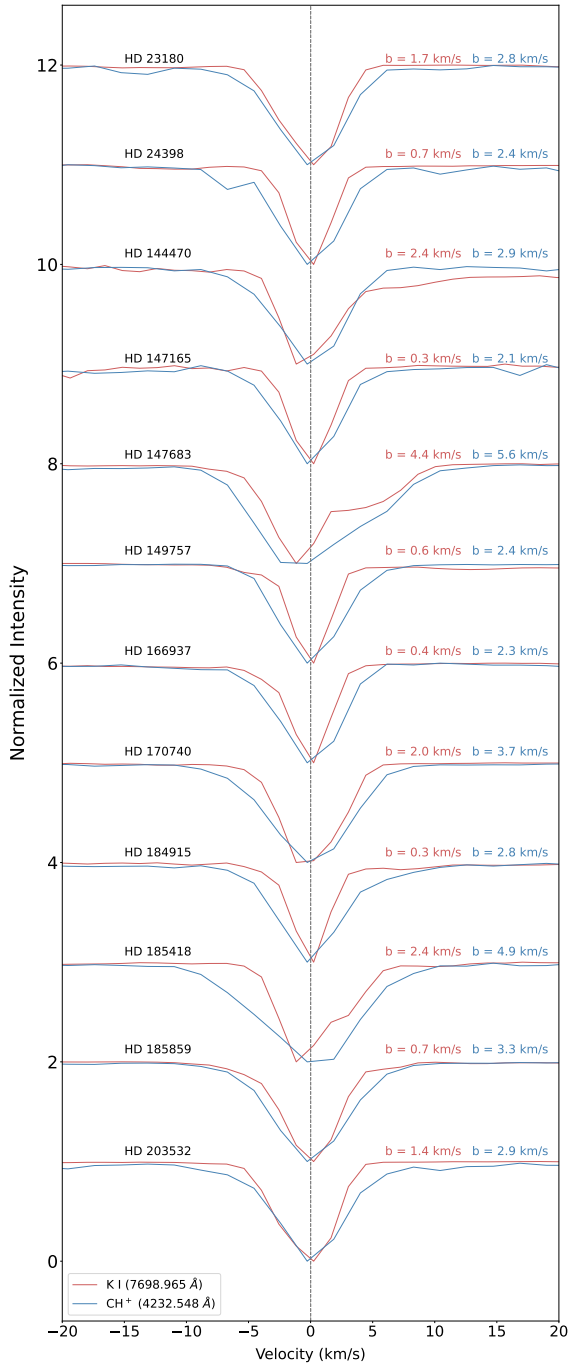


Figure 6: Profiles of KI and CH^+. The *b* parameter, which represents the gaussian width, is indicated in red for KI, and in blue for CH^+.

to a hot band[32] or due to a blend of two rotational contours peaking at 6613.6 and 6614.2 Å [34]. Ebenbichler et al.[47] however argues against the λ6614 DIB to be a blend of two rotational contours caused by two different molecules because they find a very strong correlation between λ6614 -

λ6521 DIBs (band strength Pearson correlation coefficient of 0.99). It would be very unlikely for two molecules to have a blend of transitions at two wavelengths. These questions are important to consider when discussing possible carrier molecules. We assume here that the red wing of the profile is fairly independent of the detailed substructures.

An exciting and key new finding in our work is that variations in the λ6614 DIB profile cannot be attributed solely to changes in rotational temperature, but contain a significant contribution from variations in the line width, ranging from 0.14 to 0.21 cm^{-1}. Such line widths are larger than the instrumental resolution for our EDIBLES data (σ = 0.085 cm^{-1}) and even more so for the high-resolution data from Galazutdinov et al.[38] with σ = 0.03 cm^{-1}. We can thus rule out that this is due to instrumental effects. Morerover, the 6614 DIB profile shapes are much better reproduced when including the line width as a free parameter, which strongly suggests that the observed variations are indeed due to changes in the intrinsic line width. The measured line widths then correspond to velocity dispersions (b-values) of 2.7–4.1 km s^{-1} (see Table 5). This is somewhat larger than the typical velocities measured in interstellar atomic lines e.g. $b \sim 0.2 - 2$ km s^{-1} for K I; see[48]. However, our intrinsic line widths are in the range of velocities that has been established for interstellar CH and especially CH$^+$ lines towards μ Sgr b values in the range of 1–4 km s^{-1}; see e.g.[45].

To further investigate this association, we measured the line widths of the CH$^+$ line at 4232Å by fitting a Voigt profile to the EDIBLES observations of our sightlines in that wavelength range. As a reference, we did the same for the K I line at 7699Å (see Fig. 6). Within our sample, we find that the line widths (b-values) for K I are in the range 0.3–4.4 km s^{-1} (with an average of 1.4 ± 0.3 km s^{-1}) whereas for CH$^+$, they span the range 2–5 km s^{-1} (with an average of 3.2±0.3 km s^{-1}). Thus, the intrinsic line widths we need to reproduce the 6614 DIB profile are comparable in value to the line widths of the CH$^+$ lines, but there is more: we find a good and positive correlation (with a Pearson correlation coefficient of 0.66, see Fig. 7) between the measured CH$^+$ line widths and the inferred intrinsic line widths of the 6614 DIB. On the other hand, there is no clear correlation (correlation coefficient of 0.37) with the line widths of the the K I line.

The clear correspondence between the line widths for the 6614 DIB carrier and the CH$^+$ lines

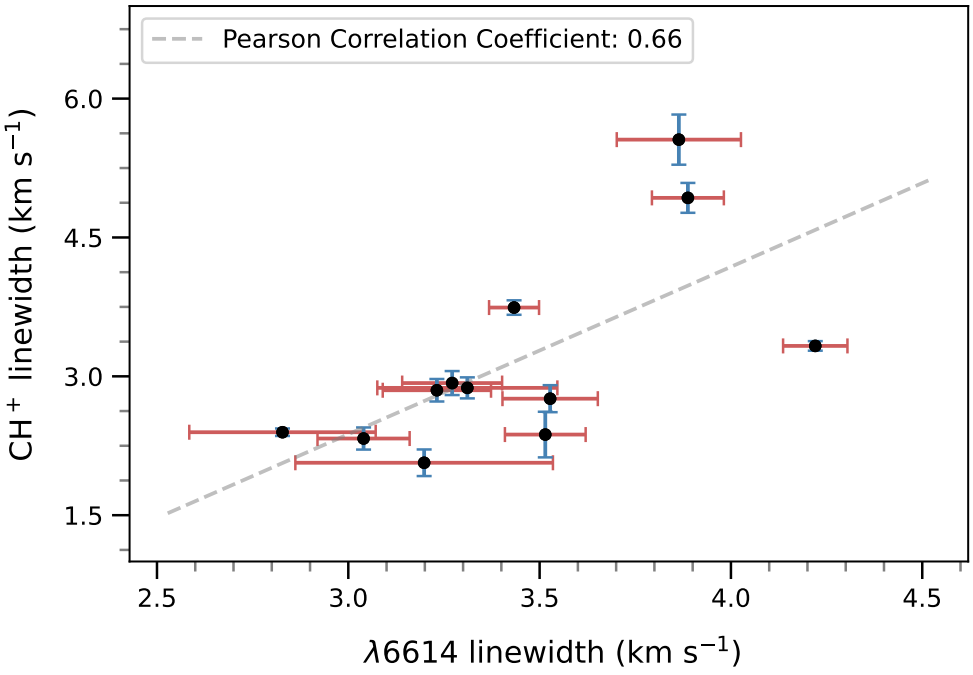


Figure 7: Linewidth of λ6614 DIB (in kms^{-1} as function of linewidth of CH^+)

indicates that the same physical processes that determine the CH^+ line widths also affect the variations in the 6614 DIB profile, and that they are thus most likely co-spatial. Recent work by Godard et al.[49] has shown that the production of CH^+ is the consequence of mass exchanges between the Cold Neutral Medium (CNM) and the Warm Neutral Medium (WNM), induced by large-scale turbulence and thermal instability. As a result, almost all of the mass in CH^+ originates from warm and unstable gas that is produced by the evaporation of CNM clouds. This process injects out-of-equilibrium H_2 into warmer and more diffuse environments where it can react with C^+ to form CH^+. The CH^+ thus resides in a low-density gas ($0.6 < n_H < 10$ cm^{-3}) with a high kinetic temperature ($600 < T < 8000$ K) and a low molecular fraction $2 \times 10^{-3} < f_{H_2} < 0.4$; [49]. The line width for CH^+ is determined by both the high thermal velocity dispersion in this hot gas and micro-turbulence. We note however that the λ6614 DIB carrier is probably a large molecule; hence thermal broadening should not play a major role for this species. The correlation between the line widths then suggests that microturbulence is in fact the key contribution to the line width for both species. We can thus conclude that also the λ6614 DIB carrier resides in this hot, low-density gas

with a low molecular fraction. It is interesting to note that λ5780 DIB has been observed towards low-density, high-temperature regions like the local bubble[50]. We note that the rotational temperatures we derive for the λ6614 DIB carrier ($T_{\rm rot}$ ~85 K) are expected to be very different from the kinetic temperature since they are determined by photo-physics rather than collisional processes.

There may be more to this correspondence in line width than just two molecular species being co-spatial. It could well be that the association of the λ6614 DIB with CH^+ has a chemical origin as well, for instance, if the λ6614 DIB carrier too is the result of non-equilibrium chemistry involving unstable H_2. The importance of vibrationally excited H_2 for interstellar chemistry was recently underscored by the detection of CH_3^+[51]; it remains to be seen what the effect is on the chemistry involving larger molecules. Thus, to gain more insights into the carrier of λ6614 DIB, we need to study in detail the chemical reactions that can occur in an environment where PAH-like molecules (including PAH radicals) are present along with vibrationally excited H_2 and C^+. These reactions might be an important formation pathway for the λ6614 DIB carrier. Additionally, since CH^+ is a transient species, we must consider whether the λ6614 DIB carrier is also transient. Linnartz et al.[52] discusses the coincidence between λ5450 DIB and absorption by a molecular transient consisting of carbon and hydrogen. Finally, it is worth pointing out that similar conclusions likely apply to the $\lambda\lambda$5797 and 6379 DIBs, as their peak separations correlate with those of the λ6614 DIB[22].

8 Summary and Conclusions

In this paper, we developed the formalism to simulate rotational contours for perpendicular transitions in oblate symmetric top molecules, and carried out a parameter study. We then used these simulated profiles to fit the profiles of the λ6614 DIB, with a focus on reproducing the profile variations that are observed across different single-cloud sightlines. We find that the λ6614 profile can be reproduced well by an oblate top containing ~54 C atoms at rotational temperatures of typically 85 K, with only a small variation of about 10 K in rotational temperature across the sightlines. Sur-

prisingly, we find that the intrinsic line width in our simulations is a key parameter to reproduce the observed profiles and their variations. The line widths are typically 3–4 km s^{-1}, and correlate well with the line widths of CH^+ lines in the same sightlines. This indicates that the λ6614 DIB carrier must reside in the same hot and low-density environment as CH^+, and could possibly be affected by chemistry involving unstable H_2.

This work shows that there is value in studying the detailed DIB profile shapes and their variations. Future work will include studying other DIBs with similar profiles (such as e.g. some of the C_2-DIBs, see Elyajouri et al.[53]), simulations for other geometries and for other types of transitions.

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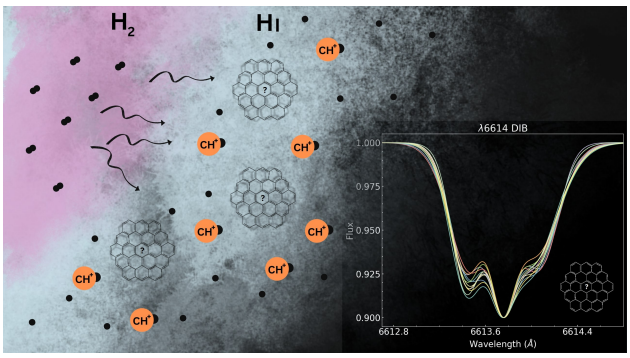
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For TOC Only


H2
HI
CH+
λ6614 DIB
1.000
0.975
0.950
0.925
0.900
Flux
6612.8
6613.6
6614.4
Wavelength (Å)

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